

SkyTeam : User Documentation

Introduction

This game is a digital version of the SkyTeam board game, brought to users using Java and libGDX. This game is a 2 Player strategy game where you and your teammate work together as a pilot and co-pilot to control a commercial airplane and get it to land on the airport runway safely. This documentation will guide you through installation, system requirements, roles, gameplay and troubleshooting.

System Requirements

Minimum Requirements

- **Operating System:** Windows, macOS, or Linux
- **Processor:** Dual-core 2.0 GHz
- **CPU Architecture:** x64
- **Memory:** 2 GB RAM
- **Graphics:** OpenGL compatible
- **Java:** JDK 17 or higher
- **Storage:** 40 MB free space

Recommended Requirements

- **Processor:** Quad-core 3.0 GHz
- **Memory:** 4 GB RAM
- **Graphics:** OpenGL 3.0 compatible
- **Java:** JDK 20 or higher
- **Storage:** 55 MB free space

Note:

Requirements stated here are to be considered after your system having JDK version and Gradle installed.

Installation

The game already comes packaged into .zip file and does not need any installation.

The .zip file can be extracted into a new folder and user can run a script depending on their Operating Systems.

Windows:

User has to run playWindows.bat file on his system by a simple double-click.

macOS or Linux:

1. Open the terminal on your system.
2. Drag and drop the SkyTeam extracted folder into the terminal. (Alternatively you can also define the folder path on your own).
3. Run the command → `./playMacOS.sh` to start the game.

Gameplay Overview

SkyTeam is a cooperative game where players take on roles to ensure the airplane lands safely. Communication and coordination are key to success.

Game Roles

- Pilot: Controls the Landing gears and brake levers exclusively.
- Co-pilot: Controls the Flaps exclusively.

All other components of the airplane can be either controlled by any player or some are controlled by both players partially.

Game Objectives

- Navigate through air traffic on the way to airport.
- Manage the airplane axis.
- Control the engine speed throughout the game.
- Land safely on the runway at the airport.

How to Play

Main menu

- **Play:** Start a new game.
- **How to Play:** Show know-hows.
- **Exit:** Close the game.

Controls

- Both Pilot and Co-pilot can control components on the airplane by dragging their dices on the coloured dice field.
- If you want to use any of the tokens during game you can click on it to activate the token.
- **Coffee Token:** Click on the token and you will see up and down arrows on your dices, you can choose to increase or decrease the value by clicking on the corresponding button.
- **Reroll Token:** Click on the Token and you will see reroll buttons on your dices, whichever player has their turn can reroll the tokens they wish to and then click on done button beside the reroll token field only once and continue playing their move. Now on next player's turn, player chooses to reroll their part of the dices and double-click on done to finally stop using reroll token and players continue with their game.

Game Rules

Players would have to setup a board and add all the components to it in an actual board game but in our digital version everything is already taken care of. As soon as you start the game, it will be on round 1 with everything on its place and dice rolled for you.

General Rules

- Players have to assign themselves a role of either of Pilot or Co-pilot and can only play on their turns.
- Dices of the current players will be showed on the screen when it is their turn and other player should not look at the dices.
- Players are allowed to discuss their strategy before each round. After that players are expected to play the game in complete silence.
- You can only place 1 dice at once on your turn and colour constraints has to be respected. Players can only place their coloured dices on the matching coloured fields.
- If placeholders are marked with specific numbers, a player can only place the corresponding value dice on that field. Players are free to place dice of any value otherwise.
- The concentration fields are marked with both colours. Therefore, any player can place the dice there.

Mandatory Actions

- You have to fill both Axis fields and Engine fields before ending the round.
- If you are out of dices and either of those 4 fields are empty, you loose the game.
- Do not forget to attend Axis and Engine fields for both players.

Reroll Token

- At any time during the round, any player may spend a Reroll token.

- This allows both players to reroll one or more of the dice behind their screen once turn by turn.

Axis

- Manage your Axis during your approach
- Axis will be adjusted after both players have placed their dice on the Axis fields.
- Do not move anything if both dice have the same number.
- If the dice show different numbers, turn the Airplane as many marks as the difference between the 2 dice.
- The Axis Arrow will turn toward the player who played the highest die.
- Axis Arrow doesn't reset itself after each round.
- If the Axis Arrow reaches or goes past X, the plane goes into a spin and you immediately lose the game.

Engine

- Depending on the power you assigned to the engines, the Airplane will advance or not.
- Plane will move forward after both players have placed their dice on the Engine fields.
- If the sum of the dices is less than or equal to the Blue marker, the plane will not advance.
- If the sum is between 2 Aerodynamics markers, approach track will be advanced by 1.
- If the sum is greater than or equal to Orange marker, approach track will advance by 2.
- If one or more Airplane tokens move into the Current Position space, you're still alive!
- If there are Airplane tokens in the Current Position space and you have to advance the Approach Track, you have had a collision, and you've lost the

game.

- If the airport is in the Current Position space and you have to advance the Approach Track, you have overshoot the airport, and you've lost the game.

Radio

- Communicate with the Control Tower to clear the traffic on your approach path.
- The Pilot has only 1 space to place a Radio die, and the Co-Pilot has 2.
- On the Approach Track, count the number of spaces, starting with your Current Position, and it'll immediately remove one Airplane token from that space. If you play a die with value 1, remove an Airplane from the Current Position space.
- This action does not have any effect if there are no Airplanes in the space indicated by the die.
- If your dice value exceed the number of positions on the approach track, it'll remove the farthest plane from the approach track.

Landing Gear (Pilot only)

- Deploy the Landing Gear. Each piece of Landing Gear deployed increases the Airplane's drag and wind resistance.
- Place a die respecting the number constraint. The order in which you deploy your Landing Gear is not important.
- Game advances the blue Aerodynamics marker one space.
- You will see a green light under the gear which is already deployed.
- Blue marker should be between the 7 and the 8 when you have deployed all the Landing Gear.
- When you move your Aerodynamics markers, you are changing your Speed's effect on the Approach Track. For example, moving the Blue marker forward means that a Speed of 5 will now advance your plane 0 instead of 1.

Flaps (Co-pilot only)

- Deploy the flaps. Each flap extended increases the aircraft's lift and wind resistance.
- Place a die respecting the number constraint. Deploy the Flaps in order, from top to bottom.
- The game advances the orange Aerodynamics marker one space. It should be just past the 12 once all the Flaps have been deployed.
- You must deploy the Flaps in order, beginning with the 1/2 space, then the 2/3 one, and so on.

Concentration

- The Pilot and Co-Pilot can place any die in any free space.
- A Coffee token will be placed in one of the 3 designated spaces. You cannot have more than 3 Coffee tokens at a time.
- Any time you place a die anywhere on the Control Panel, you can use one or more Coffee tokens to modify the value of the die you are placing. Each spent token allows you to either add 1 to, or subtract 1 from, the value of your die.
- Any player may use the Coffee tokens, regardless of who created them.
- Unspent tokens remain on the board for the following round.
- Modifiers can only change the value of a die to numbers between 1 and 6.
- Finally, subtracting 1 from a 1-value die does not change it to 6, nor does adding 1 to a 6 change it to a 1.

Brakes (Pilot only)

- Place a die respecting the number constraint. The brakes must be deployed in order, starting with the 2 space.
- The game advances the red Brake marker one space. The Brakes only have an impact in the game's final round.

Decreasing Altitude

- Regardless of what you have done during the round, the plane descends; there are 7 spaces on the Altitude Track, and therefore 7 rounds in the game.
- The game Advances the Altitude Track one space, or 1,000 feet after each round.

Special Cases —>

Reaching the Airport Too Soon

- The Airport image is in the Current Position screen but the Airplane image is not in the Current Altitude screen.
- You are in a holding pattern over the airport; you must play one (or more) rounds, without advancing any further on the Approach Track, so reduce your speed.
- If you advance the approach track now, you have overshoot the destination and you loose the game.

Not reaching the Airport in Time

- The Airport image is not in the Current Position screen, but the Airplane image is in the Current Altitude screen.
- You have crash landed before reaching the airport, you lost the game.

Final Round and End of the Game —>

- The final round begins when the Airport image appears in the Current Position screen and the Airplane image appears in the Current Altitude screen.

Victory Conditions

- There are no Airplane tokens on the Approach Track.

- All your Flaps and Landing Gear Switches show the green light.
- Your Airplane's Axis is completely horizontal.
- Your Speed is less than your Brakes when you placed your Engine dice.

Troubleshooting

Game won't start

- Make sure you have suitable version of JDK and Gradle installed on your computer.
- It can happen that your system has denied reading or writing permissions. To solve this issue please allow all the files of the game to read, write and modify in your terminal.
- **macOS and Linux:** If you cannot drag and drop your game folder in the terminal directly, than right-click on the game folder and choose an option to open that folder in terminal and then run the command given in the installation guide.

Screen Resolution Problem

- If your game elements are scattered all around the screen or you can't really see all the elements in the game window follow the steps below to set it according to your system.
- **Game Ratio:** The game is optimized to run in a **16:9 ratio**. Running in a **16:10 ratio** is possible, but the display of certain fields may look odd. Thus, it is not recommended.
- **Full-Screen Mode:** The game should run in **full-screen mode** for **Full HD** screens. For **2K or 4K** screens, it is recommended to run the game in **windowed mode**.
- **Full HD Screens:** To toggle full-screen mode, use the `setFullscreenMode()` function in the `SkyTeamGame` and `MainMenuScreen` classes.
- **2K/4K Screens:** For these screens, toggle to windowed mode by using the `setWindowedMode()` function in both the `SkyTeamGame` and `MainMenuScreen`.

- Go to the source code of the game `core/src/main/java/com/mygdx/skyteam/screens` now under this directory you will have to comment and uncomment a single line of code.

Credits

- **Development:** SkyTeam [Rathin, Marija and Truc]
- **Framework:** libGDX
- **Original Game Concept:** SkyTeam Board Game